

$$\underline{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

$$\begin{bmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} - \underbrace{\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}}_{\underline{\mu}_X}$$

$$= \underline{X} - \underline{\mu}_X$$

$$\begin{bmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \end{bmatrix} \begin{bmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \end{bmatrix}^T$$

$$\begin{bmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \end{bmatrix}^T = \begin{bmatrix} X_1 - \mu_1 & X_2 - \mu_2 \end{bmatrix}$$

$$\underline{K}_X \quad \underline{v}^T \underline{K}_X \underline{v} \geq 0 \quad \forall \underline{v} \quad \text{SEMIDEF POSITIVE} \quad \underline{v} = 0 \Rightarrow \underline{v}^T \underline{K}_X \underline{v} = 0$$

$$\underline{v}^T \in \left[\underline{Y} \quad \underline{v}^T \right] \underline{v}$$

$$\underline{Y} = \underline{X} - \underline{\mu}_X$$

$$= E \left\{ \begin{bmatrix} \underline{v}^T & 1 \end{bmatrix} \begin{bmatrix} \underline{Y} \\ 1 \end{bmatrix} \begin{bmatrix} \underline{Y}^T & 1 \end{bmatrix} \begin{bmatrix} \underline{v} \\ 1 \end{bmatrix} \right\} = E \left\{ \left(\underline{v}^T \underline{Y} \right)^2 \right\} \geq 0 \quad \underline{v}^T \underline{K}_X \underline{v} > 0 \quad \forall \underline{v} \neq 0 \quad \text{DEF. POSITIVA}$$

$$\underline{K}_X = \sum_i \lambda_i \underline{u}_i \underline{u}_i^T$$

$$\underline{v}^T \left(\sum_i \lambda_i \underline{u}_i \underline{u}_i^T \right) \underline{v} = \sum_i \lambda_i \left(\underline{v}^T \underline{u}_i \right) \left(\underline{u}_i^T \underline{v} \right)$$

$$= \sum_i \lambda_i \underbrace{\left[\underline{v}^T \underline{u}_i \right]^2}_{\geq 0} \geq 0$$

$$\underbrace{\begin{bmatrix} a_1 & a_2 & \dots & a_n \end{bmatrix}}_{\underline{a}^T} \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_n \end{bmatrix}}_{\underline{b}} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

$$\underline{b}^T \underline{a} = \begin{bmatrix} b_1 & b_2 & \dots & b_n \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \dots \\ a_n \end{bmatrix} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

$$\underline{a}^T \underline{b} = \underline{b}^T \underline{a}$$

Ass $\lambda_i < 0 \quad \underline{v} = \underline{u}_{i_0} \quad \underline{v}^T \underline{u}_{i_0} = \underline{u}_{i_0}^T \underline{u}_{i_0} = \begin{cases} A & i = i_0 \\ 0 & i \neq i_0 \end{cases} \quad \begin{matrix} A > 0 \\ \underline{A=1} \end{matrix}$

$$\sum_i \lambda_i (\underline{v}^T \underline{u}_i)^2 = \lambda_{i_0}$$

$$\underline{A} = \left(\sum_i \lambda_i \underline{u}_i \underline{u}_i^T \right) \left(\sum_e \frac{1}{\lambda_e} \underline{u}_e \underline{u}_e^T \right)$$

$$\underline{u}_i^T \underline{u}_e = \delta_{e,i} \begin{cases} 1 & e=i \\ 0 & e \neq i \end{cases}$$

$$= \sum_i \sum_e \lambda_i \frac{1}{\lambda_e} \underbrace{\underline{u}_i \underline{u}_i^T \underline{u}_e \underline{u}_e^T}_0$$

$$= \sum_i \begin{pmatrix} 1 & \\ & 0 \end{pmatrix} \underline{u}_i \underline{u}_i^T = \underline{I}$$

$$\underline{K}_x = \begin{bmatrix} \sigma_1^2 & \rho_{12} \sigma_1 \sigma_2 \\ \rho_{12} \sigma_1 \sigma_2 & \sigma_2^2 \end{bmatrix} \quad \underline{K}_x^{-1} = \frac{1}{\sigma_1^2 \sigma_2^2 (1 - \rho_{12}^2)} \begin{bmatrix} \sigma_2^2 & -\rho_{12} \sigma_1 \sigma_2 \\ -\rho_{12} \sigma_1 \sigma_2 & \sigma_1^2 \end{bmatrix}$$

$$\underline{K}_x^{-1} (\underline{x} - \underline{\mu}_x) = \frac{1}{\sigma_1^2 \sigma_2^2 (1 - \rho_{12}^2)} \begin{bmatrix} \sigma_2^2 & -\rho_{12} \sigma_1 \sigma_2 \\ -\rho_{12} \sigma_1 \sigma_2 & \sigma_1^2 \end{bmatrix} \begin{bmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{bmatrix}$$

$$= \frac{1}{\sigma_1^2 \sigma_2^2 (1 - \rho_{12}^2)} \begin{bmatrix} \sigma_2^2 (x_1 - \mu_1) - \rho_{12} \sigma_1 \sigma_2 (x_2 - \mu_2) \\ -\rho_{12} \sigma_1 \sigma_2 (x_1 - \mu_1) + \sigma_1^2 (x_2 - \mu_2) \end{bmatrix}$$

$$(\underline{x} - \underline{\mu}_x)^T \underline{K}_x^{-1} (\underline{x} - \underline{\mu}_x) = \begin{bmatrix} x_1 - \mu_1 & x_2 - \mu_2 \end{bmatrix} \frac{1}{\sigma_1^2 \sigma_2^2 (1 - \rho_{12}^2)} \begin{bmatrix} \sigma_2^2 (x_1 - \mu_1) - \rho_{12} \sigma_1 \sigma_2 (x_2 - \mu_2) \\ -\rho_{12} \sigma_1 \sigma_2 (x_1 - \mu_1) + \sigma_1^2 (x_2 - \mu_2) \end{bmatrix}$$

$$= \frac{1}{\sigma_1^2 \sigma_2^2 (1 - \rho_{12}^2)} \left\{ \sigma_2^2 (x_1 - \mu_1)^2 - \rho_{12} \sigma_1 \sigma_2 (x_1 - \mu_1) (x_2 - \mu_2) - \rho_{12} \sigma_1 \sigma_2 (x_1 - \mu_1) (x_2 - \mu_2) + \sigma_1^2 (x_2 - \mu_2)^2 \right\}$$

$$= \frac{1}{1 - \rho_{12}^2} \left\{ \frac{(x_1 - \mu_1)^2}{\sigma_1^2} - \frac{2 \rho_{12}}{\sigma_1 \sigma_2} (x_1 - \mu_1) (x_2 - \mu_2) + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} \right\}$$

$$Z = a_1 X_1 + a_2 X_2$$

$$Z_1 = a_1 x_1 + a_2 x_2$$

$$f_{X_2}(x_2) = \frac{f_{X_1, X_2}(x_1, x_2)}{f_{X_1}(x_1)}$$

$$z = a_1 x_1 + a_2 x_2$$

$$f_{z/x_1=x_2} = a_1 x_1 + a_2 x_2 \bigg|_{x_1=x_2}$$

$$f_{x_2/x_1=x_2}(x_2) = \frac{f_{x_1, x_2}(x_1, x_2)}{f_{x_1}(x_1)}$$

$$f_z(z) = \int_{-\infty}^{+\infty} f_{x_1}(x_1) f_{z/x_1=x_1}(z) dx_1$$

$$f_{x_2/x_1=x_1}(x_2) = \frac{1}{|a_2|} f_{x_2/x_1=x_1}\left(\frac{x_2}{a_2}\right)$$

$$f_{a_2 x_2/x_1=x_1 + a_1 x_1}(z) = f_{a_2 x_2/x_1=x_1}(z - a_1 x_1)$$

$$f_{z/x_1=0}$$

$$E[z] = E[Ax + b] = A E[x] + b = A \mu_x + b$$

$$z = Ax + b$$

$$z - \mu_z = (Ax + b) - (A\mu_x + b) = A(x - \mu_x) = A[x - \mu_x]$$

$$(z - \mu_z)(z - \mu_z)^T = \{A[x - \mu_x]\} \{A[x - \mu_x]\}^T$$

$$= A[x - \mu_x][x - \mu_x]^T A^T$$

$$\Sigma_z = E[(z - \mu_z)(z - \mu_z)^T] = E\left\{A[x - \mu_x][x - \mu_x]^T A^T\right\}$$

$$= A \left\{E[x - \mu_x][x - \mu_x]^T]\right\} A^T$$

$$= A \Sigma_x A^T$$

$$\begin{pmatrix} A x_1 \\ A x_2 \end{pmatrix} = A (x_1 + x_2)$$

$$\underline{b} = - \underline{A} \underline{x}_0$$

$$\underline{A} \underline{x}_1 + \underline{b}$$

$$\underline{A} (\underline{x}_1 + \underline{x}_2) + \underline{b}$$

$$\underline{A} \underline{x}_2 + \underline{b}$$

$$\underline{A} \underline{x} + \underline{b} = \underline{A} (\underline{x} - \underline{x}_0)$$

$$p_{x,y}(x,y) = p_x(x) p_{y|x}(y|x)$$

$$p_{x,y,z}(x,y,z) = p_{x,y}(x,y) p_{z|x,y}(z|x,y) = p_x(x) p_{y|x}(y|x) p_{z|x,y}(z|x,y)$$

$$p_{x_1, x_2, \dots, x_n}(x_1, x_2, \dots, x_n) = p_{x_1}(x_1) p_{x_2|x_1}(x_2|x_1) p_{x_3|x_2, x_1}(x_3|x_2, x_1) \dots p_{x_n|x_{n-1}, x_{n-2}, \dots, x_2}(x_n|x_{n-1}, x_{n-2}, \dots, x_2)$$

$$\text{rand} \in (0, 1)$$

$$\text{rand}[1, N]$$

$$\begin{bmatrix} x(1) & x(2) & \dots & x(N) \\ y(1) & y(1) & \dots & y(N) \end{bmatrix}$$

$$x(n)$$

$$\begin{bmatrix} n_1 = 5 & n_2 = 9 & n_2 - n_1 \end{bmatrix}$$

$$\begin{bmatrix} n_1 = 50 & n_2 = 54 & n_2 - n_1 \end{bmatrix}$$

$$\begin{bmatrix} n_1 & n_2 & n_3 \\ 5 & 9 & 10 \end{bmatrix}$$

$$h=45 \quad 50 \quad 54 \quad 55$$

$$\begin{bmatrix} z(1) & z(1) & \dots & z(N) \end{bmatrix}$$

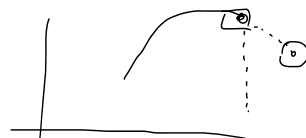
$$\begin{matrix} n_1 & n_2 & \longrightarrow & n_2 - n_1 \\ n_1 + h & n_2 + h & \longrightarrow & n_2 - n_1 \quad \forall h \end{matrix}$$

$$n_1 \quad n_2 \quad \dots \quad n_N$$

$$n_1 + h \quad n_2 + h \quad n_3 + h \quad \forall h$$

20 per cento

15 per cento



5 "1"
 3 "2"
 2 "3"

$$X_1 = \begin{cases} 1 & \frac{1}{2} \\ 2 & \frac{3}{10} \\ 3 & \frac{2}{10} \end{cases}$$

X_2	X_1			
	1	2	3	
1	$\frac{1}{2} \cdot \frac{4}{9}$	$\frac{1}{2} \cdot \frac{3}{9}$	$\frac{1}{2} \cdot \frac{2}{9}$	$\frac{1}{2}$
2	$\frac{3}{10} \cdot \frac{5}{9}$	$\frac{3}{10} \cdot \frac{2}{9}$	$\frac{3}{10} \cdot \frac{2}{9}$	
3	$\frac{2}{10} \cdot \frac{5}{9}$	$\frac{2}{10} \cdot \frac{3}{9}$	$\frac{2}{10} \cdot \frac{1}{9}$	

$$\left(\frac{4}{9} + \frac{3}{9} + \frac{2}{9} \right) = \frac{1}{2}$$

$$\left(\frac{5}{9} + \frac{2}{9} + \frac{2}{9} \right) = \frac{3}{10}$$

$$P[X_2=1 | X_1=2] = \frac{5}{9}$$

$$P[X_1=1, X_2=1] = \underbrace{P[X_1=1]}_{\frac{1}{2}} \underbrace{P[X_2=1 | X_1=1]}_{\frac{4}{9}}$$

$$P[X_1=2 | X_1=1] = \frac{3}{9}$$